Linux Server Administrator Exam Report

Question

An institute looking to establish a secure and efficient Linux server environment for database management must follow several key steps. Begin by setting up an LDAP server to centralize user authentication. Then, deploy PostgreSQL with LDAP-based authentication to ensure secure database access. Next, configure Joomla to be securely accessible over HTTPS on port 443 using a designated domain name. Additionally, implement a file-sharing service to manage user home directories via FTP with LDAP authentication, and set up an SSH server to authenticate users through LDAP.

Server Configurations

OS: - Ubuntu-Server-24.04.1
IPv4: - 192.168.1.7
Hostname: - armour.local
Domains: - armour.local, www.armour.local

Credentials

OS Users:

Users	Password
root	Yashjoshi@8796
Ubuntu	Yashjoshi@8796

SSH Users:

Users	Password
Ubuntu	Yashjoshi@8796

Client Configuration

OS: - Debian
IPv4: - 192.168.1.6

Credentials

Users	Password
root	Yashjoshi@8796
Armour	Yashjoshi@8796

LDAP Configuration

Credentials

Users	Password
infosec	Yashjoshi@8796
anony	Yashjoshi@8796

Login Server with SSH Service

└─$ ssh ubuntu@192.168.1.7

Configure a DNS server on Ubuntu Server OS using BIND9.

1- Install BIND9 DNS Server

```
# sudo apt update
# sudo apt install bind9 bind9utils bind9-doc
```

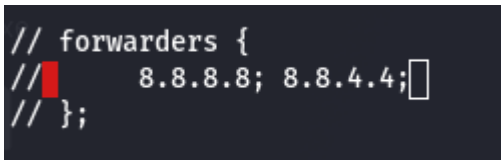
2-Configure BIND9

Edit the main BIND configuration file:

```
# sudo nano /etc/bind/named.conf.options
```

Uncomment and modify:

```
forwarders {
8.8.8.8; 8.8.4.4;
};
```



save and exit

3- Create a Zone File

Edit the local zone configuration:

 # sudo nano /etc/bind/named.conf.local

zone "armour.lcoal" {

type master;

file "/etc/bind/forward.armour.local";

};

```
zone "armour.lcoal" {
    type master;
    file "/etc/bind/forward.armour.local";
};
```

save and exit

Create the DNS Zone Database File

Copy a template:

 # sudo cp /etc/bind/db.local /etc/bind/forward.armour.local

 # sudo nano /etc/bind/forward.armour.local

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     ldap.armour.local. root.armour.local. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@               IN      NS      ldap.armour.local.
ldap            IN      A       192.168.1.7
armour.local.   IN      A       192.168.1.7
www             IN      CNAME   armour.local.
client          IN      A       192.168.1.8
~
~
```

save and exit

5. Check Configuration and Restart BIND

```
# sudo named-checkconf
```

 # sudo named-checkzone armour.local /etc/bind/forward.armour.local

If no errors:

```
# sudo systemctl restart bind9
```

 # sudo systemctl enable named-resolvconf.service

now add dns server in client

 # vim /etc/resolv.conf

```
└─# cat /etc/resolv.conf
nameserver 192.168.1.7
```

Now Add rules in firewall.

Now resolve dns with dig command

```
┌──(root㉿debian-server)-[/home/armour]
└─# dig ldap.armour.local.

; <<>> DiG 9.18.33-1~deb12u2-Debian <<>> ldap.armour.local.
;; global options: +cmd
;; Got answer:
;; WARNING: .local is reserved for Multicast DNS
;; You are currently testing what happens when an mDNS query is leaked to DNS
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 22123
;; flags: qr aa rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 66af537eba5891110100000067b1f510eed885f2813b1529 (good)
;; QUESTION SECTION:
;ldap.armour.local.             IN      A

;; ANSWER SECTION:
ldap.armour.local.      604800  IN      A       192.168.1.7

;; Query time: 0 msec
;; SERVER: 192.168.1.7#53(192.168.1.7) (UDP)
;; WHEN: Sun Feb 16 19:54:16 IST 2025
;; MSG SIZE  rcvd: 90
```

Now Configure LDAP Server

1. Install OpenLDAP Server and Utilities

```
# sudo apt update
# sudo apt install slapd ldap-utils
```

During the installation, set an LDAP admin password.

2. Reconfigure slapd (if needed)

If not prompted, run:

```
# sudo dpkg-reconfigure slapd
```

- **No to omitting the LDAP server configuration.**
- **Enter your domain name (e.g., example.com).**
- **Enter the organization name.**
- **Set admin password.**
- **Select MDB as the database.**
- **Yes to removing the database on slapd purge.**
- **Yes to move the old database.**

3. Verify LDAP Installation

```
# sudo systemctl status slapd
```

You should see it running.

5. Create LDAP Base Structure

Create a file named `base.ldif`:

```
# sudo nano base.ldif
```

dn: ou=users,dc=armour,dc=local

objectClass: organizationalUnit

ou: users

dn: ou=groups,dc=armour,dc=local

objectClass: organizationalUnit

ou: groups

```
ubuntu@ubuntu-server:~$ cat base.ldif
dn: ou=users,dc=armour,dc=local
objectClass: organizationalUnit
ou: users

dn: ou=groups,dc=armour,dc=local
objectClass: organizationalUnit
ou: groups
```

Apply this:

 # ldapadd -x -D "cn=admin,dc=armour,dc=local" -W -f base.ldif

You can generate a password hash with:

`slappasswd`

{SSHA}RQzJNmJX1gI7va40s1KCrPw1pr7HA/dS

6. Add a User to LDAP

Create infosec_user`.ldif`:

`# sudo nano infosec_user.ldif`

dn: uid=infosec,ou=users,dc=armour,dc=local

objectClass: inetOrgPerson

objectClass: posixAccount

objectClass: shadowAccount

cn: Infosec Armour

sn: Armour

uid: infosec

uidNumber: 10000

gidNumber: 10000

homeDirectory: /home/infosec

loginShell: /bin/bash

userPassword: {SSHA}RQzJNmJX1gI7va40s1KCrPw1pr7HA/dS

```
ubuntu@ubuntu-server:~$ cat infosec_user.ldif
dn: uid=infosec,ou=users,dc=armour,dc=local
objectClass: inetOrgPerson
objectClass: posixAccount
objectClass: shadowAccount
cn: Armour Infosec
sn: Infosec
uid: infosec
uidNumber: 10001
gidNumber: 10001
homeDirectory: /home/infosec
loginShell: /bin/bash
userPassword: {SSHA}SR8PhlnTx7YHEV+FcMV38QcEs5rVppJd
```

Add the user:

 # ldapadd -x -D "cn=admin,dc=armour,dc=local" -W -f infosec.ldif

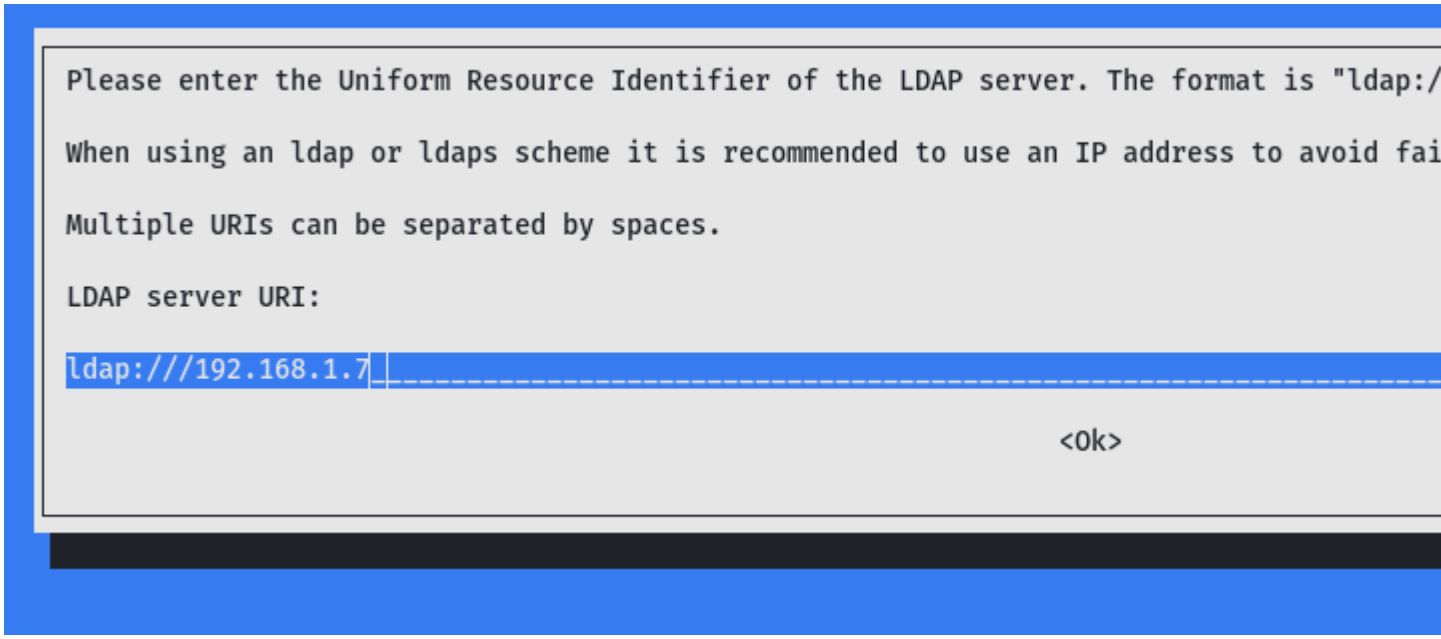
7. Configure Client to Authenticate with LDAP

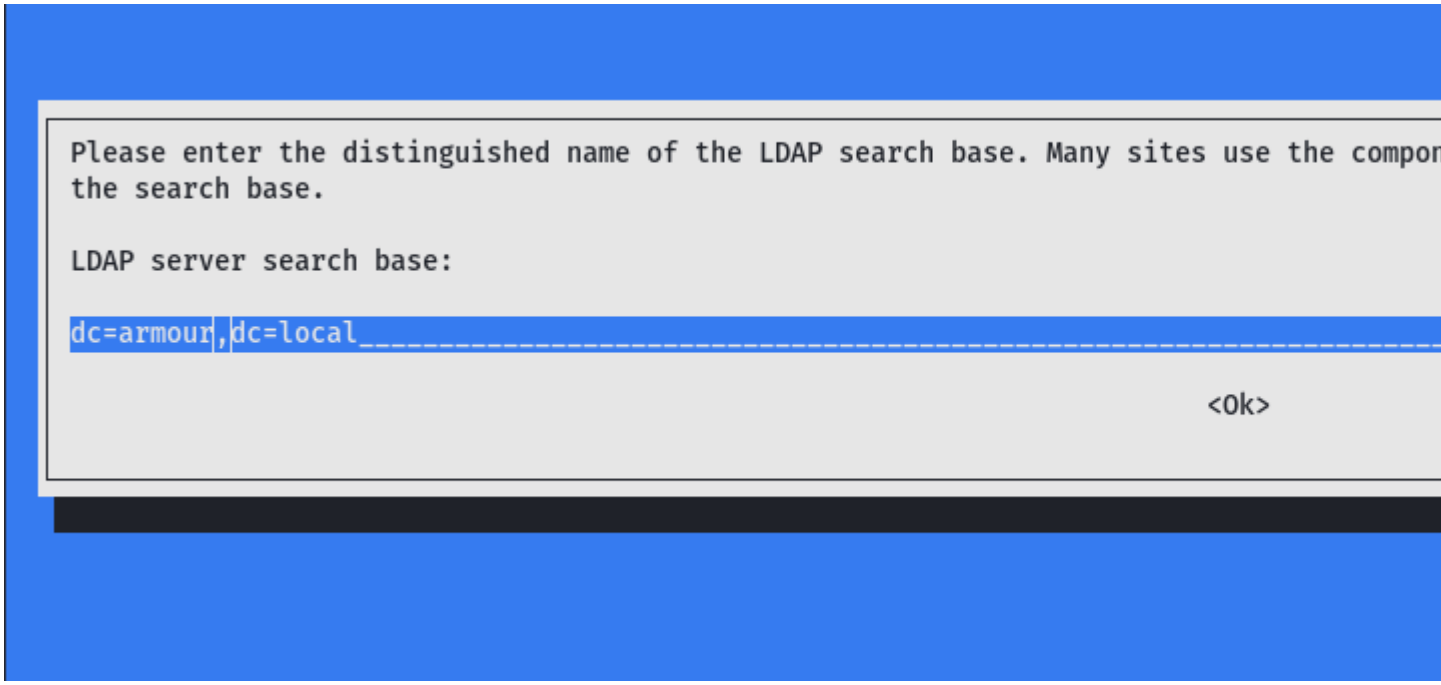
On the client machine, install necessary packages:

```
# sudo apt install libnss-ldap libpam-ldap ldap-utils nscd
```

During installation, provide:

- **LDAP Server URI: ldap://192.168.1.7**
- **Base DN: dc=armour,dc=local**
- **LDAP version: 3**
- **Make local root user admin? Yes**
- **Does LDAP require login? No (unless configured)**

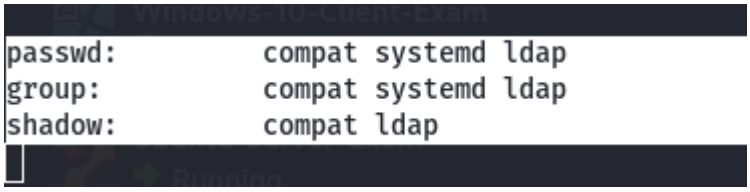




8. Configure NSS for LDAP Authentication

Edit `/etc/nsswitch.conf`:

```
# sudo nano /etc/nsswitch.conf
```



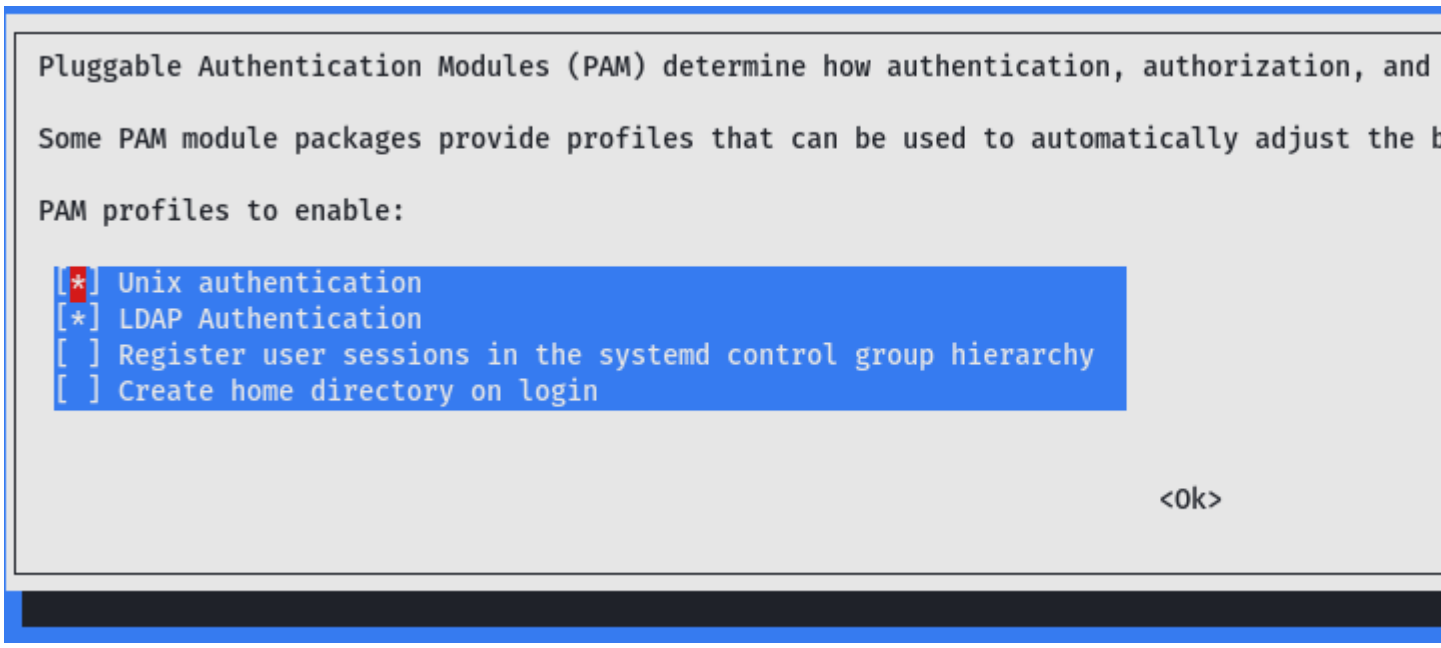
9. Configure PAM for LDAP Authentication

Ensure PAM modules are enabled:

```
# sudo pam-auth-update
```

Select:

- **Unix authentication**
- **LDAP authentication**



10. Check Firewall on LDAP Server

Ensure the firewall on the LDAP server allows traffic on port 389 (LDAP) or 636 (LDAPS).

On the LDAP server:

```
sudo ufw allow 389
sudo ufw allow 636
sudo ufw reload
```

11. Update LDAP Authentication Files

Edit `/etc/ldap.conf` or `/etc/ldap/ldap.conf`:

```
# sudo nano /etc/ldap/ldap.conf
```

Ensure these lines exist:

uri ldap://192.168.1.7

base dc=armour,dc=local

ldap_version 3

binddn cn=admin,dc=armour,dc=local

bindpw Yashjoshi@8796

pam_password md5

```
#
# LDAP Defaults
#

# See ldap.conf(5) for details
# This file should be world readable but not world writable.

#BASE    dc=example,dc=com
#URI     ldap://ldap.example.com ldap://ldap-provider.example.com:666

uri ldap://192.168.1.7
base dc=armour,dc=local
ldap_version 3
binddn cn=admin,dc=armour,dc=local
bindpw Yashjoshi@8796
pam_password md5

#SIZELIMIT      12
#TIMELIMIT      15
#DEREF          never

# TLS certificates (needed for GnuTLS)
TLS_CACERT      /etc/ssl/certs/ca-certificates.crt
```

```
# sudo systemctl restart nscd
# sudo systemctl restart nslcd
```

12. Test LDAP Client

Verify with:

 # getent passwd infosec

output- infosec:x:10001:10001:Armour Infosec:/home/infosec:/bin/bash

You should see LDAP user info.

Step 5: Test LDAP Authentication Try switching user:

 # `su -` infosec

Configure Joomla for secure HTTPS access on port 443 with a designated domain name.

Install Joomla on Ubuntu Server

 # sudo apt update

 # sudo apt install apache2 mysql-server php php-mysql libapache2-mod-php php-xml php-mbstring php-zip php-curl php-gd php-intl unzip

Download Joomla:

 # wget https://update.joomla.org/releases/5.2.3/Joomla_5.2.3-Stable-Full_Package.zip

Unzip and move to `/var/www/html/joomla`:

 # sudo unzip Joomla_5.2.3-Stable-Full_Package.zip -d /var/www/html/joomla

Set proper permissions:

 # sudo chown -R www-data:www-data /var/www/html/joomla

 # sudo chmod -R 755 /var/www/html/joomla

2. Configure Apache for Joomla

Create a new Apache virtual host config:

#sudo nano /etc/apache2/sites-available/joomla.conf

<VirtualHost *:80>

ServerName armour.local

DocumentRoot /var/www/html/joomla

Redirect permanent / https://armour.local/

</VirtualHost>

<VirtualHost *:443>

ServerName armour.local

DocumentRoot /var/www/html/joomla

SSLEngine on

SSLCertificateFile /etc/ssl/armour.local/armour.local.crt

SSLCertificateKeyFile /etc/ssl/armour.local/armour.local.key

<Directory /var/www/html/joomla>

AllowOverride All

</Directory>

</VirtualHost>

save and exit

Enable the new site and required Apache modules:

 # sudo a2ensite joomla

 # sudo a2enmod ssl rewrite

3. Obtain SSL Certificate

 # openssl req -x509 -nodes -days 365 -newkey rsa:2048 -keyout /etc/ssl/armour.local/armour.local.key -out /etc/ssl/armour.local/armour.local.crt

5. Restart Services

 # sudo systemctl restart apache2

1. **Access Joomla Web Installer**
2. **Open a browser and go to:https://armour.local**

2. Joomla Web Installation Wizard

1. **Language Selection: Choose your preferred language.**
2. **Site Name: Enter the name of your site (e.g., `Armour Local`).**
3. **Admin Account: Create an admin account by providing:◇ Username**
4. **Password**
5. **Admin Email**

Click Next.

3. Database Configuration

1. **Database Type: PostgreSQL**
2. **Host Name: `armour.local`**
3. **Username: Your PostgreSQL user (e.g., `root` or a Joomla-specific user).**
4. **Password: The PostgreSQL users password.**
5. **Database Name: Name of your Joomla database (e.g., `joomla`).**
6. **Table Prefix: Automatically generated (e.g., `joo_`).**

Click Next.

4. Final Setup

1. **Review settings.**
2. **Install Sample Data if needed.**
3. **Click Install to finalize the Joomla setup.**

5. Enable HTTPS in Joomla

Once installed:

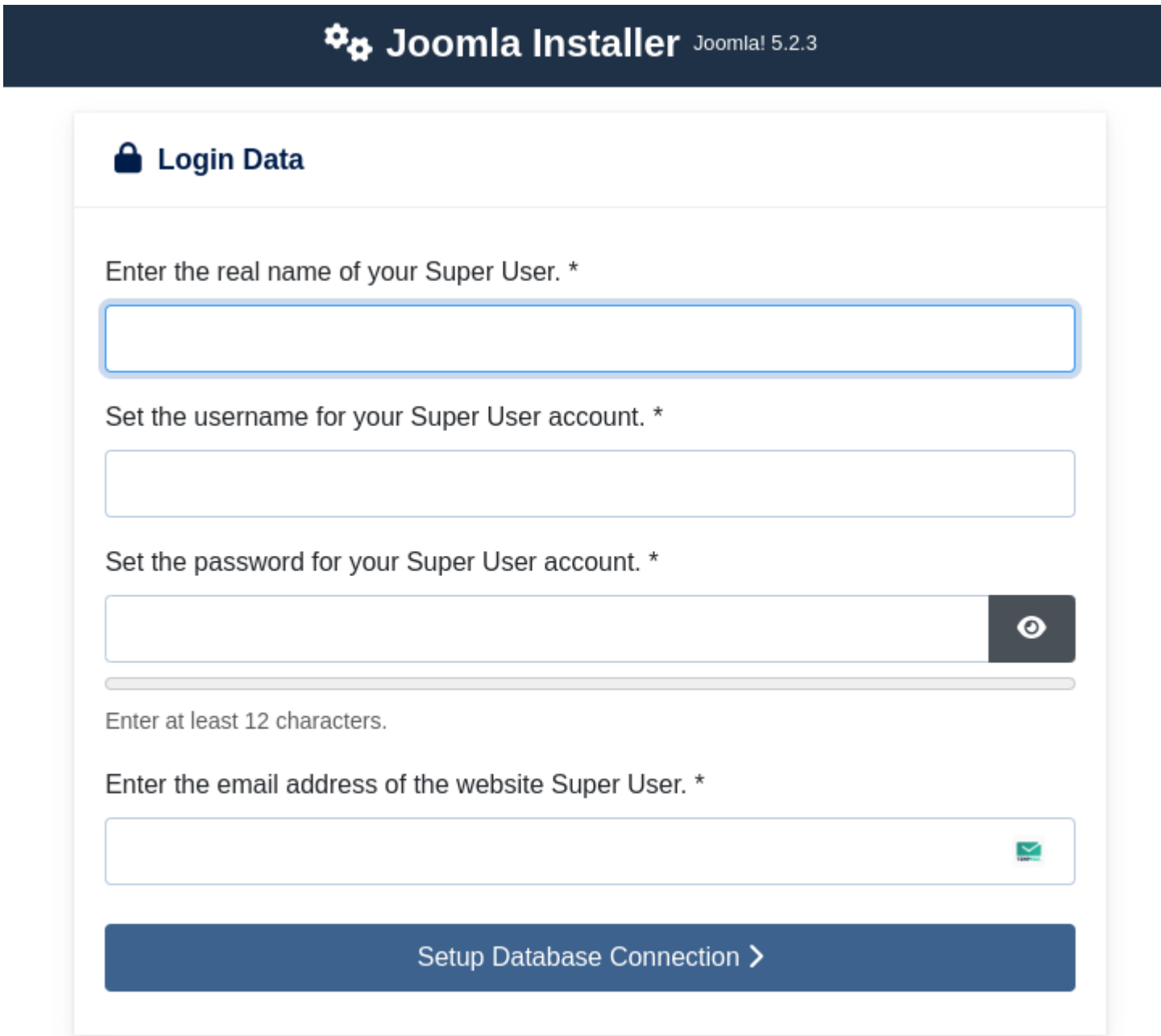
- **Log in to the Joomla Admin Panel at:https://armour.local/administrator**
- **Go to System > Global Configuration.**
- **Set Force HTTPS to Entire Site.**
- **Save and close.**

6. Adjust Apache Permissions

 # sudo chown -R www-data:www-data /var/www/html/joomla

 # sudo chmod -R 755 /var/www/html/joomla

 # sudo ufw allow 443



2. Joomla Web Installation Wizard

1. **Language Selection: Choose your preferred language.**
2. **Site Name: Enter the name of your site (e.g., `Armour Local`).**
3. **Admin Account: Create an admin account by providing:◇ Username**
4. **Password**
5. **Admin Email**

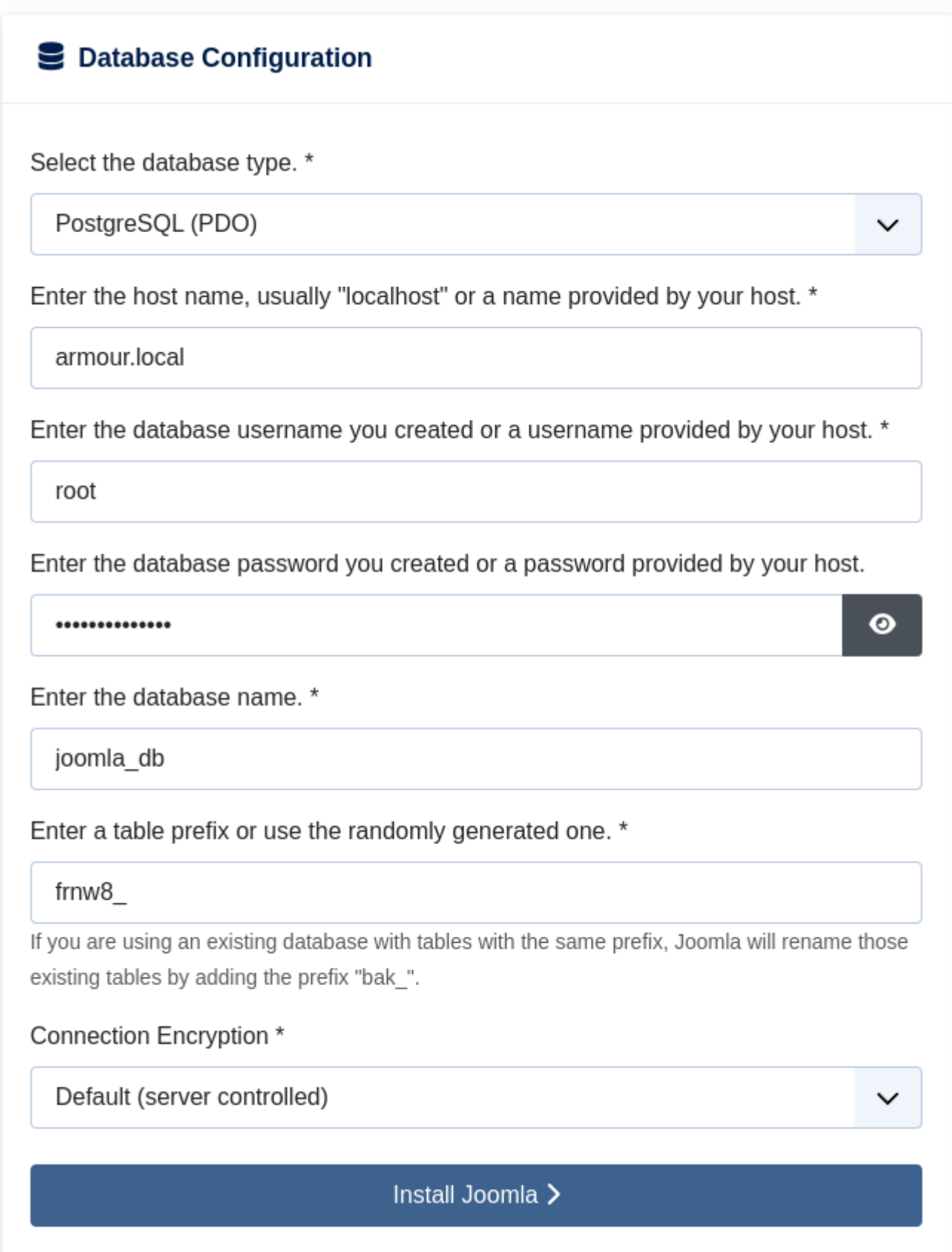
Click Next.

3. Database Configuration

1. **Database Type: PostgreSQL**
2. **Host Name: `armour.local`**
3. **Username: Your PostgreSQL user (e.g., `root` or a Joomla-specific user).**

4. **Password: The PostgreSQL users password.**
5. **Database Name: Name of your Joomla database (e.g., `joomla`).**
6. **Table Prefix: Automatically generated (e.g., `joo_`).**

Click Next.



Now Configure vsftpd for FTP

1. Install vsftpd

 # sudo apt update

 # sudo apt install vsftpd libpam-ldapd curlftpfs -y

2. Configure vsftpd

Edit the `vsftpd` configuration file:

 # sudo nano /etc/vsftpd.conf

Add or modify the following lines:

listen=YES

listen_ipv6=NO

anonymous_enable=NO

local_enable=YES

write_enable=YES

chroot_local_user=YES

allow_writeable_chroot=YES

pam_service_name=vsftpd

ssl_enable=NO

Explanation:

- **`local_enable=YES`: Allows local users (LDAP users) to log in.**
- **`write_enable=YES`: Allows file uploads and modifications.**
- **`chroot_local_user=YES`: Restricts users to their home directories.**
- **`pam_service_name=vsftpd`: Tells vsftpd to use PAM for authentication.**

5. Restart vsftpd

 # sudo ufw allow ftp

```
# sudo systemctl restart vsftpd
```

6. Verify FTP Access from Server

Try logging in via FTP from the server to verify:

```
# ftp 192.168.1.7
give LDAP username and password
```

Configure SSH Services

Step 1: Install Required Packages

First, install the necessary packages for LDAP and SSH configuration on your server.

 # sudo apt update

 # sudo apt install libnss-ldap libpam-ldap ldap-utils ssh

Configure NSS (Name Service Switch) to Use LDAP: Edit `/etc/nsswitch.conf` to include `ldap` for `passwd`, `group`, and `shadow`.

 # sudo nano /etc/nsswitch.conf

passwd: compat ldap

group: compat ldap

shadow: compat ldap

Configure PAM (Pluggable Authentication Modules): Edit the `/etc/pam.d/common-auth` file to configure LDAP authentication.

 # sudo nano /etc/pam.d/common-auth

auth required pam_ldap.so

 # sudo nano /etc/pam.d/sshd

auth required pam_ldap.so

Step 3: Configure SSHD to Use PAM

Ensure that the SSH server uses PAM for authentication.

1. **Open the SSH configuration file.**

 # sudo nano /etc/ssh/sshd_config

- **Find the `UsePAM` directive and set it to `yes`.**

```
UsePAM yes
```

- **Restart the SSH service to apply changes.**

```
# sudo systemctl restart ssh
```

Step 4: Test LDAP Authentication

Test LDAP authentication by attempting to SSH into the server using an LDAP user.

```
# ssh infosec@192.168.1.7
```

```
ubuntu@ubuntu-server:/home$ ssh infosec@192.168.1.7
infosec@192.168.1.7's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-53-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Feb 17 03:55:29 PM UTC 2025

  System load:              0.22
  Usage of /:               31.2% of 23.45GB
  Memory usage:             17%
  Swap usage:               0%
  Processes:                171
  Users logged in:          1
  IPv4 address for enp0s3: 192.168.1.7
  IPv6 address for enp0s3: 2401:4900:1c18:74e2:f1e6:fd04:db74:a969
  IPv6 address for enp0s3: 2401:4900:1c18:74e2:aca1:8c25:4b61:c5f3

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

131 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Mon Feb 17 15:55:30 2025 from 192.168.1.6
groups: cannot find name for group ID 10001
infosec@ubuntu-server:~$
```